

Audio Feedback in Gaming: How Audio Properties Shape Player Experience

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Abstract

This study investigates how audio feedback in video games influences player experience, with a focus on four specific audio properties (Pitch, Loudness, ADSR Envelope, and Frequency Content) across three classic interactions: collecting an item, making a UI action, and taking damage. The study consisted of two phases, the data gathering phase and the audio analysis phase. The first phase involved the player test, where players chose the sounds to be investigated in the second phase. The resulting sounds were extracted and analyzed according to the four audio properties. The analysis indicated, among other things, that UI action sounds should be short, noise-like clicks to effectively respond to the player's actions. Item collection sounds should also be fast, but lean towards being rewarding and positive, while the taking damage sounds should be slightly longer and lean towards being punishing, impactful, and negative. Audio feedback remains a relatively underexplored area in game design research, and these findings—along with potential future ones should this paper be expanded upon—provide more scientifically and technically grounded guidelines for sound designers and game developers looking to give players more effective feedback in their games.

Keywords

Interaction design, sound design, video games, audio analysis, immersion

1 Introduction

Since the rise of the commercial video game industry in the early 1970s, video games have become increasingly popular and established themselves as a central part of global society [18]. Even though most people might not consider themselves gamers, most have probably played some type of video game recently. With this in mind, understanding what makes video games enjoyable is important at every level. What actually makes a video game enjoyable depends heavily on the type of video game. For the purposes of this paper, however, an extremely broad generalization can be made. A video game comprises the visuals—which include the graphics, animation, 3D models, textures, and lighting in the game—and the audio—which includes audio feedback, ambient noise, cues, sound effects, and music. Most existing research on this topic focuses on the visuals, not the audio [11, 15]. Therefore, this paper focuses on the audio half of video games.

Audio is a significant part of video games. It helps the player in a wide variety of ways, including guiding the player to make correct

decisions with audio cues [16] and enhancing the player's experience with immersive sound [8]. This paper specifically examines audio feedback.

Audio feedback is a relatively straightforward concept. It can essentially be viewed as a fundamental function of human-computer interaction (HCI). When an action is performed by a human through interaction with a computer, the computer consequently sends confirmation that it has received the action by providing the human with some type of audio in return. In the case of video games, however, the human becomes a player, and the computer becomes a game.

With a relatively concrete definition of audio feedback in place, the possibility of observing different audio feedback situations (AFS) opens up. It is important to determine what makes an AFS satisfying for the player, as it is generally subjective and differs depending on the player and the AFS in question. Consider an AFS in a first-person shooter (FPS) action game, and then consider the playful UI and gameplay of Candy Crush Saga [17]. If one were to choose a random AFS from each of these video games and then task a random test player with evaluating both AFSs, this would not yield any useful results. The test invites a plethora of sources of error, such as the player having a bias towards their preferred game or the player being familiar with one of the games but having no idea what the other game's AFSs should feel like.

Player tests are still relevant since players are the ones playing the games—and therefore should decide what is good or bad. However, controlling subjectivity and sources of error in the tests beforehand is highly important. In this paper, this is done partly by asking participants specific questions before they take part in the test, but also by focusing on similar types of AFSs from different games instead of different AFSs from similar games. The term classic interaction is used in reference to similar interactions that exist in many video games. The classic interactions that are used in this paper are the following:

- (1) Collecting an item (coins, pickups, loot),
- (2) Making a UI action (clicking a button, accepting a menu option),
- (3) Taking damage.

1.1 Objective

The objective of this paper is to determine how audio properties of sounds from classic interactions in audio feedback situations influence a satisfactory player experience in video gaming. The results can be used as a guideline when implementing classic interactions—or creating new AFSs—in the development of new video games.

2 Theory

In order to properly analyze the effect of audio feedback, and determine how relevant audio properties from AFSs actually affect player experience, it is necessary to first establish a theoretical framework. This section introduces the key concepts and theories that this study relies on, including audio feedback, audio properties, player experience, and the previously mentioned AFS concept.

2.1 Audio Feedback

Audio feedback should, as stated, be viewed as a fundamental function of HCI, defined as the use of non-verbal sound to convey information to users (players) in a computer interface [12].

Since this paper explores audio feedback in video games, pivoting from an HCI-focused view to a more gaming-centered view is important. In video games, audio feedback is not just a tool for conveying information—it serves as a medium through which players can interact with the game [8], making it much more versatile and broader than in a strictly HCI-focused definition.

To make this clearer: imagine a study on users’ perceived levels of focus while performing a task through a computer interface. Users would answer ten easy questions twice—once with no sound and once with a positive *ding* indicating a correct answer. The idea is that this study would most likely yield obvious results: that focus is better with the *ding* as a motivator. However, this is not the case in video games, because audio feedback’s purpose there is not as binary. In video games, audio feedback is not meant to do one thing, but many different things depending on what action the player is taking.

Therefore, in this context, audio feedback can be understood as any sound generated in direct response to a player’s input or the game’s own internal events or actions, with the purpose of informing, guiding, or immersing the player. This expanded definition forms the basis of the analysis conducted in this paper.

2.2 Audio Properties

Audio properties, as a concept, come from the study of acoustics and include amplitude, frequency, wavelength, and velocity [9, 20]. These are the fundamentals of what sound is and how it is physically perceived by the human ear. They are analogous, meaning they need to be manipulated or converted to be represented digitally [10], and therefore do not necessarily make sense to examine in this paper since audio is not handled in its analog form.

Consequently, a set of audio properties far more relevant to this study has been chosen. These are:

- Pitch,
- Loudness (dB),
- ADSR Envelope,
- Frequency content.

2.2.1 Pitch. Pitch is a primary auditory sensation [20], and is admittedly its own concept regardless of modality. It is nonetheless selected as an audio property due to it being central to human auditory perception. One might think that pitch and frequency content are similar properties to examine. However, when examining the pitch of a sound, you are examining the sound’s tonality, the actual tone of the sound. But when you examine the frequency

content, you are also looking at the resonant frequencies of the sound. Therefore, by examining both, a more holistic analysis of the sound can be established.

2.2.2 Loudness (dB). Loudness is, along with pitch, considered a primary auditory sensation [20]. It refers to the term used to describe the magnitude of an auditory sensation as perceived by the listener [7]. In physics, this refers to the actual amplitude of a sound; that is, the degree to which the sound wave oscillates. This is true in this context as well; however, a more digital and measurable approach for the actual analysis is still preferable. Therefore, a more suitable definition of loudness in this case is the sound pressure level measured in decibels (dB).

2.2.3 ADSR Envelope. ADSR Envelope refers to the model used to describe a sound’s shape over time [21]. ADSR is an acronym that stands for Attack, Decay, Sustain, and Release. These are the four stages that determine how a sound evolves from the moment it starts to the moment it ends.

A sound’s attack refers to the time it takes for a sound to reach its peak loudness. For example, a short burst of sound, like a quick *beep*, will have a very short attack because there is virtually no gradient from the moment the sound is triggered until its peak can be observed. The decay is the time it takes for the sound to move from its peak loudness to its sustained level. The sustain is simply the level at which the sound continues to play while it is triggered. Lastly, the release describes the time it takes for the sound to completely dissipate to silence after it stops being triggered.

The ADSR Envelope is mainly a term used in electronic music and music production [21]. However, it is still a highly relevant metric for this paper due to the nature of the sounds under investigation, since the ADSR Envelope describes a sound over time—a characteristic that directly corresponds to how a player might perceive the sound in real time when playing a game. See Figure 1 for an example of how the shape of the ADSR Envelope influences the shape of the waveform of a sound.

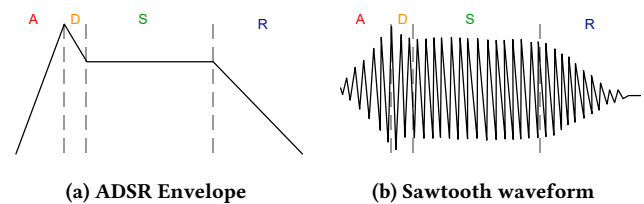


Figure 1: Example of how an ADSR Envelope shapes the waveform of a sound.

2.2.4 Frequency content. The final audio property under investigation in this paper is frequency content. Frequency content, in this paper, refers to the full spectral composition of a sound as represented on the frequency spectrum [9]. Frequency content could, as stated, be considered a metric similar to the pitch of a sound. However, examining the entire frequency spectrum of a sound yields more insights than simply examining only the pitch. For example, if a melodic sound is examined—that is a sound that has a tone—this sound has a root note that is the pitch. But the sound most likely has

some resonant frequencies [9] that further determine its character. These resonant frequencies are the ones referred to as frequency content.

2.3 Audio Feedback Situation (AFS)

Another concept that is important to understand is the Audio Feedback Situation, or AFS. AFS refers to a specific type of situation in which a player finds themselves while playing a video game. In this situation, the player receives a sound from the game as a direct response to an input, with the purpose of informing, guiding, or immersing the player, which consequently allows the player to act accordingly. This type of situation is often rather short because the sound that conveys the information from the game to the player needs to be concise and correspond correctly to the game's state.

The difference between AFSs and audio feedback is that AFSs are applied. This means that there is a need for context to fully grasp the concept. For example, imagine an AFS where a player attacks an enemy in a video game with a sword. The player makes an input by swinging their sword towards an enemy, and the enemy (the game) in turn produces a sound of pain, indicating to the player that their swing hit. It is this type of situation or scene that defines what an AFS is.

To further define this, consider another example of an AFS that places significant responsibility on the interaction within the context; that is, the player making a UI action. Clicking a button to confirm settings, for example. In this AFS, the player would most certainly be left confused if they changed their settings and went to confirm them, only to be met with nothing. The game needs to give the player appropriate audio feedback—such as a *click*—to make it clear that the new settings have been confirmed.

2.4 Player Experience

Player Experience refers to the overall subjective response that a player has while playing a video game [1]. This includes perceptions, emotions, and cognitive dimensions. This means it is a multidimensional concept that requires careful scoping in the context of this paper.

Since this paper focuses on audio feedback, player experience was primarily examined through how audio influenced the player's perception and responses to the game world. Therefore, the aspects of player experience that were most relevant in this paper are the ones directly affected by audio. These aspects included immersion, emotional response, and the sense of presence within the game [3].

Furthermore, because player experience is inherently subjective, the way players are asked to evaluate the AFSs also becomes important. Consider the AFS where a player swings their sword at an enemy once again. There is a difference between asking the player taking the test: "Which version do you prefer?" and "Which version makes you feel most like the character swinging the sword?" It is important that the player is given enough context about what to consider when choosing an alternative, and that the choice is somewhat connected to the aspects of the player experience defined here.

It is important to note, however, that player experience is not necessarily what is under scrutiny in this paper, that is the audio properties that influence player experience the most. The reason player experience is important to grasp is to establish a more grounded understanding of the problem at hand.

3 Method

The method of this study consisted of two main phases. That is the data gathering phase and the audio analysis phase. The overarching goals of the two phases were such that the first phase featured player tests in which players were tasked with picking the sounds that were analyzed in the second phase. Then, in the second phase, the analysis of the relevant sounds was performed with respect to the four audio properties previously mentioned. Based on this analysis, the results were presented, and a proposition aligned with the study's objective was offered.

3.1 Data Gathering Phase

The core of the data gathering phase was to perform player tests in which players were presented with three AFSs from each classic interaction. The players chose which sound they found best suited the AFSs and proceeded until they had chosen a preferred sound for each of the nine AFSs. There were two versions of each AFS, one with the original sound from the actual game and one with an alternative version of the sound. The different sound could be either a slightly manipulated version of the original sound or a different sound that is similar to the original. The two versions were short videos that properly mediated the AFS. The structure of the player tests is visualized in Figure 2. The goal of the player tests in the data gathering phase was to let actual players determine which sounds make an AFS satisfactory so that there would be a legitimate basis on which an analysis could be conducted.

Recall the three classic interactions mentioned in the introduction. These interactions act as a foundation for the data gathering phase because they are AFSs that are similar, regardless of whether the games they come from are similar or not. The term classic interaction is also intentional: they are classic because many games make use of similar mechanics. If different games communicated these mechanics (or interactions) to the player in different ways, the players would likely take longer to understand the game's mechanics. Furthermore, the three chosen classic interactions cover three different areas of interaction between the player and the game, which is also fitting for the study. Collecting an item covers the player's interaction with the game world, making a UI action covers the interaction between the player and the game's user interface, and taking damage corresponds to the interaction between the player and a threat.

This makes it possible to present the user with three different AFSs for each of the three classic interactions—that is, with AFSs from different games. The reason for using three AFSs per classic interaction is simply because nine sounds are a fairly reasonable amount to analyze in the audio analysis phase, considering the size of this study. Should this experiment be replicated, including more samples might be beneficial. Also, the players were asked specific questions before being presented with the two alternatives of the AFSs at hand in the test. These questions were: "Which alternative

makes the act of collecting the item feel most rewarding?" for collecting an item, "Which alternative gives you the most confidence that your action was registered and will lead to the expected outcome?" for making a UI action, and "Which alternative makes you most aware that your character just got hurt?" for taking damage. The purpose of these questions was to guide the player in the correct direction with their choice of alternative with respect to the aspects of player experience defined in Section 2.4.

3.1.1 Sources of Error, Limitations, and Countermeasures. The player tests were conducted in person or via digital meeting, with the researcher being available to the players for guidance if needed. There were a total of nine players. The environment was reasonably controlled, and the players took the test in silence, mostly so that they could focus on the audio in the player test, but also to avoid accidental distraction. The players also had the option to listen to the AFSs repeatedly. A broad focus was placed on recruiting people with experience playing video games. However, the players chosen for the test did not officially belong to any specific target audience. It could be beneficial, but was certainly not crucial, if the players taking the test had not heard the sounds beforehand. This is because the risk of the players choosing the familiar sound instead of the sound they actually think fits the AFS better is reduced. But the same thing can be said if the players do know the sounds beforehand. That way, they have an easier time connecting the sound to a feeling and basing their choice of alternative on that.

Other limitations in the data gathering phase include the choice of sounds and games in the player test. The test contained one fabricated sound, while the other sound was the actual sound from the game. Having both sounds fabricated was certainly an option; that way, it would be impossible for participants to have heard the sounds before. However, due to the relative unimportance of controlling that bias, this was an accepted trade-off, along with the fact that it made the design of the test simpler. As for the games, choosing ones that are relatively unknown would, similar to the sounds, mean that even experienced players would not recognize the sounds. Although, for the same reason as the sounds, this was also accepted as a trade-off.

Not all sources of bias in the data gathering phase have been controlled, which can be considered a limitation of this study. This is acceptable, however, as the data gathering phase is not central to the objective of this paper. Its purpose is simply a means of gathering sounds from relevant situations in video games that players genuinely prefer. Consequently, the reasons behind a player's choice matter less than the fact that the choice is grounded in the aspects of player experience discussed in this paper.

3.2 Audio Analysis Phase

Following the player tests, the nine preferred sounds were examined and analyzed according to their audio properties. These audio properties were selected carefully because they are fairly consistent across classic interactions and enable the identification of patterns in which sounds players prefer in an AFS. Also, this phase did not include any external participants. This phase was the central part of the study; it was here that actual observations could be made and the study's objective could be addressed. This warrants a deeper and more thorough explanation of this phase.

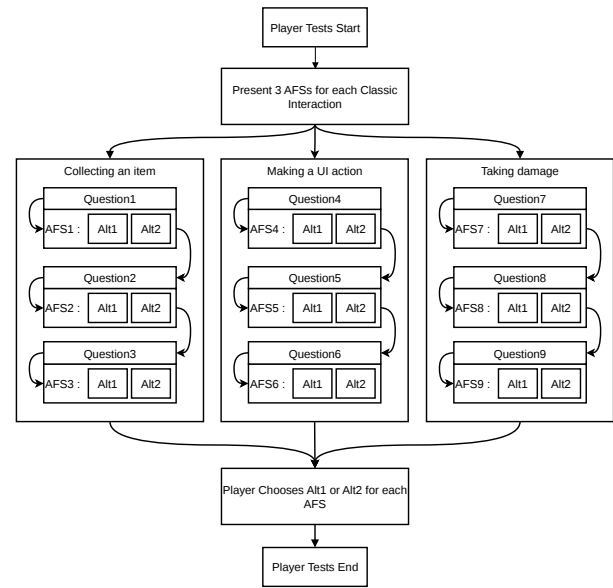


Figure 2: Flowchart illustrating the data gathering phase and the player test.

3.2.1 Setup. First, the nine sounds resulting from the player tests were collected and, if needed, converted into M4A format. This allowed for an in-depth analysis of each sound according to the four chosen audio properties. The sounds were then imported into a digital audio workstation (DAW) [22], where they were examined using several different tools and methods. The choice of DAW does not matter for this experiment, and the same analysis can be recreated in all types of DAWs, given that one has access to the same—or similar—types of tools.

3.2.2 Pitch and Frequency Content. For pitch and frequency content, the spectral components of each sound were examined using a parametric equalizer plugin called Parametric EQ 2 and an audio editing software called Edison. The equalizer made it possible to locate where on the frequency spectrum the sound was most intense, thus revealing its pitch. Edison further helped with pitch identification with its "Detect Pitch Regions" feature, which divided the sound into regions and labeled the dominant tone of each with its corresponding note.

3.2.3 Loudness. Loudness was also examined using Edison, as well as the built-in volume monitors in the DAW. It is a rather simple audio property to examine; however, it was crucial that the loudness of each sound be kept consistent between its original context from the game and the player test. If not, players might have selected AFSs not because they fit the interaction best but because they were louder than the other alternative.

3.2.4 ADSR Envelope. The final audio property under examination, the ADSR Envelope, is perhaps the most subjective property of the four. This raises a question about the methodology: should

a standard envelope shape be defined for each of the three classic interactions, against which the sounds could be compared to ensure that they work in context? For the purposes of this paper, a standard envelope shape was not established. Instead, the way the property was examined was that the waveforms of each sound were examined visually, and the observations of their appearances directly informed the results.

3.2.5 Sources of Error, Limitations, and Countermeasures. Since the analysis in the audio analysis phase was performed by a human, there are some inherent subjectivity issues regarding the interpretation of results. This is relevant for all four audio properties, but mainly the ADSR Envelope. Measures were taken to avoid this specifically for the ADSR Envelope; however, for the remaining three properties, this is still a limitation. Although pitch, frequency content, and loudness could be measured with objective tools [9], the interpretation of those measurements themselves—and what they mean in the context of an AFS—still relied on the researcher’s subjectivity. What this means is simply that while the results are reproducible, the conclusions might differ slightly from researcher to researcher.

Furthermore, regarding the digital formatting of the sounds, there are some areas worth mentioning when it comes to limitations. Mainly, the M4A format’s compression artifacts and the potential loss of quality when handling the sounds digitally. It was, because of this, a conscious choice to use M4A instead of MP3 due to M4A’s superior compression standards [4].

Aside from this, perhaps the two largest limitations of this study are the single analyst problem and the limited sample size. Although the limited sample size is a limitation, it is not critical—it only affects the generalization of the conclusions and not their validity. The conclusions remain meaningful within the scope of the nine analyzed sounds, and the limited sample size simply invites further research on the topic to confirm whether the observed patterns persist at larger sample sizes. The single analyst problem is likely more critical, however. It is well established [19] that qualitative analyses benefit from second reviewers to strengthen the reliability of the results. Unfortunately, this paper does not include a second reviewer or an inter-rater reliability check, which also is a limitation of the analysis.

4 Results

Of the nine resulting sounds from the data gathering phase, seven of them were the original sounds and two of them were alternative versions. See Figure 3 for the sound choice distribution by the players for each of the three classic interactions. This analysis yielded several findings, and the basic discoveries from the analysis are shown in Table 1.

4.1 Pitch and Tonality

Regarding pitch and tonality, a pattern emerged across the interactions. The UI sounds were almost entirely non-tonal, which aligns with their short and clicky character. However, the item sounds and the damage sounds were more varied, with some sounds having tonality and some not. This suggested that functional or negative interactions leaned toward noise-based sounds, while rewarding interactions were more accepting of tonality.

4.2 Frequency Content

The analysis indicated that the frequency content clearly separates the interaction types. For example, the UI sounds seemed to be wide in the frequency spectrum, mostly resembling short bursts of white noise. However, the item sounds were more present in the mid to high ranges, often having more pronounced peaks, giving the sounds a more pleasant feeling. The damage sounds seemed to slightly favor the lower end of the spectrum, with a low to mid emphasis.

4.3 Loudness (dB)

This section of the results is perhaps the most straightforward of the four properties. It was clear that loudness is dependent on context. Here, the damage sounds stood out in that they were, in general, louder than the UI and item sounds. Loudness also seemed to scale with the situation’s significance, so taking damage—which is an important event in a video game—needed to have a louder sound to convey its urgency.

4.4 ADSR Envelope

The ADSR Envelope findings are perhaps the most notable, as they map relatively cleanly directly onto player experience. Regarding the UI actions, they all had a near instantaneous attack, short or no sustain, and a quick decay/release. This suggested that instant confirmation and responsiveness were important for UI actions. The item sounds generally had a fast attack; however, they had a longer decay/release, and the damage sounds had a slightly slower attack with a similar decay/release to the item sounds. The two latter interactions also mapped to player experience surprisingly well. The item sounds’ long tails could be experienced as rewarding, which is beneficial for item collection, and the damage sounds’ similar tails, paired with their longer attack, communicated a more menacing character, and caused the feedback to linger and emphasize the impact.

4.5 Summary

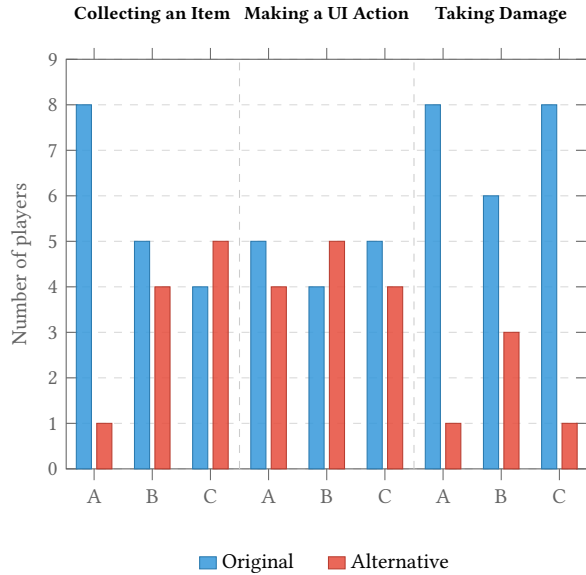
For all four audio properties, the classic interactions showed consistent patterns that align with their roles in player experience. In general, the UI action sound should be short and broad frequency-wise, giving an effective response to the player. For the item sounds and the damage sounds, there was an interesting contrast: they tended to be quite similar for most properties, but shifted towards positivity for the item sounds, and negativity for the damage sounds. For example, the damage sounds tended to be lower frequency-wise than the item sounds.

5 Discussion

The objective of this paper was to investigate how audio properties of sounds from classic interactions in AFSs influence a satisfactory player experience in video gaming. The audio analysis in this study yielded findings that support this objective—that sounds from classic interactions do indeed follow consistent patterns that correspond to each interaction and how they relate to player experience. This section discusses these findings and connects them to the study’s objective and theoretical framework.

Table 1: Results from data gathering phase and the basic discoveries from the analysis. A = Attack, D = Decay, S = Sustain, R = Release.

Sound	Frequency Content	Pitch/Tonality	Loudness (dB)	ADSR Envelope
Collecting an item A	Peak, 400 Hz	No tonality	-23 dB	Fast A, immediate D, long S, long R
Collecting an item B	No peak, 3500–13000 Hz	No tonality	-36 dB	Very fast A/D, short S, medium R
Collecting an item C	Peaks, 1850, 2600, 6120 Hz	A	-34 dB	Slow into fast A, slow D, no S, R = D
Making a UI Action A	No peak, 400–12000 Hz	No tonality	-22 dB	Instant A, instant D, no S, long R (reverb)
Making a UI Action B	S1: 1350 Hz; S2: 13500 Hz	S1: F#; S2: none	-38 dB	S1: short A, rapid D, no S, long R; S2: noise-like, very short
Making a UI Action C	S1: 500 Hz; S2: 6000 Hz	S1: none; S2: none	-39 dB	S1: layered, short; S2: click, instant A, no D/S, immediate R
Taking damage A	No peak, 170–19500 Hz	No tonality	-13 dB	Moderate A, immediate D, no S, slow R
Taking damage B	Peaks, 400, 900, 1300 Hz	A	-21 dB	Moderate A, some S, slow R
Taking damage C	Peaks, 600, 2000 Hz	E	-35 dB	Quick A, slow D, no S, short R

**Figure 3: Player test results showing the sound choice distribution. Total participants: $n = 9$.**

There are four concepts that make up this study's theoretical framework; see Section 2. These are audio feedback, audio properties, player experience, and AFSs. Perhaps the most obvious connection between the findings and the theory is how the results relate to player experience. It was stated in Section 2.4 that player experience

was "examined through how audio influences the player's perception and responses to the game world." This was exactly what the ADSR Envelope findings showed. For example, item sounds with a long release connect to immersion and reward because they give the player a lingering feeling of success [6, 13]. Damage sounds, however, linger similarly but might have a slower attack, which makes them feel more menacing.

Naturally, the ADSR Envelope is not the only concept from the theory that relates to the findings. The other audio properties also proved to be relevant. It makes sense that the UI sounds—which only need to convey confirmation, not a positive or negative emotional response—do not need tonality to be effective. For the other two types of interactions, however, the results suggested that pitch and frequency content played a part in what type of emotion the sound was trying to convey. This aligns with previous findings on how acoustic cues communicate emotional expression [14].

The results for the loudness are also interesting, despite being more susceptible to fallacies due to the difficulty of normalizing levels beforehand. The results in this case showed that loudness was dependent on context and scaled with the situation's significance, which is consistent with previous studies showing that auditory urgency is indeed influenced by loudness, or intensity [5]. However, this was harder to confirm due to the fact that a sound could be louder compared to another sound because of differences in their game's contexts.

Aside from the audio properties and their relation to player experience, there is a necessity for the other concepts from the theory as well. However, this might not relate as much to the results and conclusions of the study itself, as much of it is meant to define and clarify contexts. For example, a concrete definition of audio

feedback, as found in Section 2.1, is necessary in order to understand what the findings represent. Additionally, the definition of the abbreviation AFS is meant to simplify and specify discussions on this particular topic, giving future researchers the option to use it in the same way if this study is expanded upon or if a similar idea is explored. That said, should this paper receive a follow-up study, it is recommended to include more than three sounds for each classic interaction and to introduce a second analyst to increase reliability [19]. It might also be worth revising the four audio properties and exchanging loudness for something more relevant, such as duration, for example. Loudness yielded interesting results, but there is too much uncertainty surrounding it, which makes it too hard to draw reliable conclusions from.

Lastly, the findings are generalizations of the results, and the patterns discussed here are products of that. Therefore, the limitations should be considered when reading or replicating the study, or when using the results as a design guideline. In other words, the findings should be understood as interpretations grounded in theory rather than definitive conclusions [2].

6 Conclusions

Sound design and audio feedback are relatively unexplored areas in video gaming. This paper set out to present a more scientifically and technically grounded understanding of what audio properties players actually prefer in different AFSs from games.

The objective was to investigate how audio properties of sounds from classic interactions in AFSs influence a satisfactory player experience in video gaming. Based on the findings, it can be concluded that all three classic interactions showed clear signs of patterns that directly correspond to the theoretical framework in the paper. More specifically, the ADSR Envelope showed a direct relation to player experience regarding neutral, negative, or positive emotional responses. The pitch and frequency content also showed similar patterns in that they mattered for what type of emotional response was desired. Loudness, on the other hand, was slightly less consistent. This does not concern the results themselves, which also showed patterns, but rather about how difficult it was to confirm the observations. There are several ways this study could be improved upon. Namely, by exchanging loudness as an audio property for a more suitable metric, increasing the number of analysts, and expanding the number of sounds for every interaction. This will further strengthen the reliability of the findings.

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